

Ganesh Chandrasekar

AI/ML Engineer | NLP | LLMs | RAG | Information Retrieval

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PROFESSIONAL SUMMARY

AI/ML engineer with research and software engineering experience across NLP, retrieval, RAG, and LLM systems. Built large-scale biomedical QA pipelines, contributed to TREC and BioCreative systems, and turned thesis research into a live local-first QA product.

TECHNICAL SKILLS

Programming: Python, SQL, TypeScript/JavaScript, Java

ML & NLP: PyTorch, Hugging Face Transformers, scikit-learn, BERT, CRF, Sequence Labeling, NER, PPO

Retrieval & LLMs: RAG, BM25, Pyserini/Lucene, MedCPT, FAISS, Reranking, vLLM, Ollama, LLM Evaluation

Software & Cloud: REST APIs, Flask, Node.js, Docker, GCP, AWS, Linux, Slurm, Git

PROFESSIONAL EXPERIENCE

Research Assistant

CLaC Lab, Concordia University

Sep 2023 – Present

Montreal, Canada

- Built retrieval and QA pipelines over **26.8M PubMed documents** using BM25, MedCPT, cross-encoder reranking, and evidence selection.
- Ran quantized Qwen inference on GPU-based HPC using **vLLM** and Slurm for large-scale generation and evaluation.
- Developed evaluation workflows to separate retrieval, answer quality, grounding, and citation failures, contributing to TREC and BioCreative shared-task systems.
- Trained BERT-family, CRF, hierarchical, and PPO models for sequence labeling, and built NER pipelines across biomedical and news text.

Teaching Assistant

Concordia University

Sep 2024 – May 2026

Montreal, Canada

- Led tutorials and office hours for Information Retrieval and Web Search (COMP 479/6791) and Discrete Mathematics (COMP 232), supporting assignments, grading, and technical questions.

Solution Developer

OrangeScape Technologies Pvt. Ltd.

Jul 2022 – Aug 2023

Chennai, India

- Developed procurement workflows and product features for Kissflow Procurement Cloud across purchasing, approvals, supplier management, and source-to-pay processes.
- Built API services on **Google Cloud Platform (GCP)** and translated business requirements into workflow rules, validations, and approval paths with backend and product teams.

Software Engineering Intern

Basik Marketing Pvt. Ltd.

May 2022 – Jul 2022

Chennai, India

- Developed a real-time esports broadcast overlay with Python and Node.js, including backend logic for live match and event counters.

PROJECTS

BioEvidence AI: Local-First Biomedical Question Answering

[Live Demo ↗](#)

RAG / Pyserini/Lucene / Local LLMs / Next.js

- Built BioEvidence from my biomedical QA research, combining Pyserini/Lucene retrieval over a **14.9M-record PubMed index**, adaptive reranking, and local LLM generation.
- Implemented a real-time query-to-citation pipeline with on-device inference and integrations for PubMed/NCBI, Europe PMC, and ClinicalTrials.gov.

Knowledge-Graph and Hybrid Retrieval for Course Enrollment

[GitHub ↗](#)

RDF/RDFS / SPARQL / Apache Fuseki / LangChain / FAISS

- Built a Concordia course-enrollment knowledge base from CSV/JSON data, representing courses, lectures, topics, students, and grades as **43,291 RDF triples** across **5,258 course URIs**.
- Implemented question answering over Apache Fuseki with dynamic SPARQL queries, then compared graph, FAISS vector, and hybrid retrieval for natural-language enrollment questions.

Agentic and Non-Agentic Multi-Hop Medical QA

[GitHub ↗](#)

RAG / smolagents / Qwen2.5 / Wikipedia / PubMed

- Implemented the agentic Wikipedia workflow and compared it with controlled Wikipedia and Wikipedia + PubMed QA pipelines.
- The Wikipedia + PubMed pipeline reached **0.67 development-set accuracy**, the best result among the evaluated configurations.

Biomedical Negation Detection

[GitHub ↗](#)

PyTorch / BERT / BioBERT / CRF / PPO

- Fine-tuned BERT, BioBERT, and ClinicalBERT for cue/scope BIO tagging, comparing CRF and hierarchical variants designed around class imbalance.
- Trained a PPO-based sequence-labeling policy from annotated spans, improving minority-label detection over the supervised baseline.

EDUCATION

Master of Computer Science (Thesis)

Sep 2023 – Aug 2026

Concordia University

Montreal, Canada

Thesis: *Evidence-Grounded Biomedical Question Answering: Design, Analysis, and Evaluation*

Defended August 2026.

Bachelor of Engineering in Computer Science and Engineering

Jun 2018 – Apr 2022

KCG College of Technology, Anna University

Chennai, India

CGPA: 8.81/10, First Class with Distinction

SELECTED PUBLICATIONS

TREC 2025 BioGen | *Modular Sparse and Dense Retrieval for Evidence-Constrained Biomedical Question Answering*

BioCreative IX, IJCAI 2025 | *Agentic and Non-Agentic Multi-Hop Systems for Medical Question Answering*